Grower Data QA/QC
Supporting Document `IndigoCarbon_US-1_2024_0032`

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1 Ensuring accuracy and correctness of quality grower data

The purpose of this document is to describe the process by which the Carbon data pipeline collects grower data (Gate 1), validates and assures data quality (Gate 2), completes evidence review (Gate 3), and, as needed, does grower outreach to correct or collect additional information (Gate 4). Gate 2 is designed to put safeguards in place for the use of gapfilling with extrapolation, as well as trigger the use of gapfilling with default values to fill or replace data (in Gate 5 of the Carbon data pipeline). Together, these gates ensure grower data meets SEP v1.1 requirements, such that grower data are compliant for use in emission reduction modeling (Gate 6 of the Carbon data pipeline) and credit quantification (Gate 7 of the Carbon data pipeline).

Organization and gate structure The following quality assurance, quality control (QA/QC) workflow outlines how the management practice data that flow through Gates 1 – 4 of the Carbon data pipeline are entered, stored, reviewed, confirmed, and transformed to support the carbon accreditation process. While we collect grower data on other management activities, this document only applies to project activities detailed in Section 3.1.1 of the Monitoring Plan.

The QA/QC workflow stops the data at various checkpoints (i.e., gates) where they are reviewed and confirmed for accuracy and quality. Any data changes that occur at these checkpoints are stored as new timestamped snapshots, so data provenance and reproducibility is conserved. Growers will iterate through these checkpoints as they enter new and update existing data. Each checkpoint increases in complexity and is built upon successful completion of

the previous. Fig. 6.7. in the Monitoring Plan gives a view of the overall checkpoint structure, with information contained herein focusing on Gates 1 – 4.

1.1 Gate 1: Basic user interface validation

Gate 1 includes basic validation checks associated with the data collection user interfaces of Carbon by Indigo (used by growers and Indigo field staff). Information that does not pass these checks will not be allowed in the system *at all*. These basic validations include (but are not limited to) correct data types (e.g., float, int, string, date, etc.), precision of a variable, number is non-negative, practice change matches accepted list, selection from dropdowns, etc., and required attributes must be entered before saving is allowed. When a grower believes their data are complete, they can request to have their data reviewed by Indigo. Upon approval, the grower can submit to the Project via the "final filing" process.

Final filing is the process by which growers confirm all of their entered information and submit them to Indigo for processing (similar to how one might file their yearly income taxes). During final filing they attest to having reviewed all data and the information in this submission, and then agree that everything is complete and accurate to the best of their knowledge. At this point, their fields are locked and no other changes can be made (especially to information that comprises their baselines).

As noted in the Monitoring Plan, management data are entered into the Carbon program platform mainly through the Carbon by Indigo software. Management data are structured in terms of "events" that encompass typical field activities of a cropping season. An (entire) event is considered a *qualitative* piece of data (e.g., yes/no that a fertilizer was applied), and within each event are both *qualitative* and *quantitative* data (e.g., whether or not a fertilizer application included a nitrogen inhibitor vs fertilizer application rate, respectively). The grower survey and management data are organized into eight core events:

1. Cash Crop Planting,
2. Cover Crop Planting,
3. Tillage,
4. Chemical Applications (with non-fertilizer chemical options),
5. Harvest and Termination/ Failure,
6. Organic Amendments (including manure),
7. Irrigation, and
8. Grazing

As noted above, events capture associated *qualitative* and *quantitative* data termed "attributes" (e.g., date, rate, equipment, etc.). Each event is organized by the date on which it occurred on a field. Each event can be saved as occurring on the full field or only on a portion of the field, with the option for the grower to draw the subfield region (also called an "event geometry"). Finally, the grower can attach evidence (i.e., corroborating proof) to each event, as needed. For more information on events, associated attributes, and their options/ formats, see IndigoCarbon_US-1_2023_0029.

In addition to manual data entry, the Carbon by Indigo software includes data ingestion from third-party Farm Management Systems (FMS): Corteva's Granular Insights, Verdova, and MyJohnDeere (MJD). API connections were built to import boundaries, machine data events, and manually entered FMS events from Granular Insights FMS. Granular Insights users were also able to import farm records that were recorded manually throughout the year in Granular's interface, which they could then import into Carbon by Indigo without needing to duplicatively re-enter their records in the Carbon by Indigo UI. A similar approach was taken to import machine records from Verdova FMS software, which includes machine data from John Deere and Climate Fieldview. Additionally, through a series of integration APIs, we directly connect to a grower's MJD cloud account and ingest their John Deere boundaries and recorded data, such as as-planted, as-harvested, and as-applied fertilizer events. These accounts store machine readings and sensor data taken by equipment, such as tractors, planters, combines, and sprayers. These integrations allow growers to select their fields and management data for import into Carbon by Indigo, saving them hours of manual data entry while also bolstering the integrity of their data in some cases, as machine data are considered historical management records (i.e., 1a in Section 6.1 of the SEP v1.1). This information is integrated with our field

boundary and event geometry constructs and, as such, follows the same Carbon data pipeline as manually entered data; they are only differ in their distinction among the hierarchy of data sources as will be discussed below in Gate 3.

1.2 Gate 2 : Grower data completeness & confidence

Once the growers have entered their data, we can pass field and event information through the grower data validation service: a series of checkpoints (6) to scrutinize the completeness and quality of this information. The validation service evaluates all aspects of the data across the following categories.

- **Inclusion** – A grower can participate in a carbon project because **Indigo can process and use** their management practice data and information.
- **Quality** – A grower can participate in a carbon project because all provided management details **follow the data source hierarchy.**

The validation service is composed of six levels (shown in the following table), each building on the complexity of the previous. Note that the type of check is consistent with the terminology noted above, and the level represents the relative scrutiny of the evaluation – the higher the level, the more in-depth the checks. The most expansive of these checkpoints is Level 4 (data confidence).

Table 1. Gates and levels in the validation service in ascending order.

Gate 2 Level or Turnstile	Description of Gate 2 Level or Turnstile	Type of Assurance Check
Level 1	Event Establishment	Quality
Level 2	Conflict Event Logic	Quality
Level 3	Practice Change Assessment	Quality
Level 4	Data Confidence	Quality
Turnstile 2	Compares attributes to remote sensing results	Quality
Turnstile 3	Compares attributes to third-party data	Quality
Turnstile 4	Compares attributes to agronomic data from Indigo subject matter experts (SMEs)	Quality
Turnstile 5	Compares attributes to agronomically feasible values	Quality
Level 5	Additionality Assessment	Inclusion
Level 6	Gapfillability	Inclusion

ECHO environmental check Per Section 3.6 of the SEP v1.1, we make meaningful effort to ensure acres in the Project are not in violation of environmental regulations. While we explicitly ask growers about these violations in the user interface, we also select a random subset for manual review using the Environmental Protection Agency's (EPA's) Enforcement and Compliance History Online (ECHO) web app. (For more information on ECHO, please see IndigoCarbon_US-1_2023_0069.)

Selecting entities for review As part of the ECHO check, we determine if an entity should be randomly selected for manual EPA ECHO follow-up review. The likelihood that an entity will be selected is based on characteristics of the fields within that entity, namely: if a given field's total acreage is above or below the threshold. In other words, sampling is done *per field*, and we do N random draws with a defined sampling rate for an entity with N enrolled fields. Table 5 demonstrates the relationship between entity sampling ratio and this criterion. The output of this sampling is the centroid of enrolled fields, which is the key by which we will search for entities using the EPA ECHO web app.

Table 5. Entity sampling ratio for manual ECHO follow-up, based on entity's field characteristics.

Entity's field characteristics	Sampling Rate
Total acres $<=$ 250 ac	10%
Total acres $>$ 250 ac	15%

Reviewing selected entities with ECHO Once an entity has been randomly selected for manual EPA ECHO follow-up review, the Field Investigator applies the following standard operating procedure to evaluate if an entity passes or fails. The objective of this process is to identify if any environmental violations noted by the EPA ECHO web app would prohibit a grower or field from participating in our program. We need to search for participating growers and confirm if they have any violations listed. If there are violations listed, then we follow up and determine if those violations relate to management practices. If so, those fields are excluded.

For each grower (entity) on the list, we use the advanced search option, and search by the latitude and longitude of the centroid of the operation determined by finding the centroid of all the operation's enrolled fields.

In the compliance and enforcement section of the search options, we remove the filter on "any violation," so the search returns *any* violation while ignoring facilities without issues. We also adjust the radius around each centroid to 50 miles to ensure we capture all possible locations for the entity.

Once the search returns, we review any results for any entities that match the name of the grower to determine if that grower has had a violation. We look for intuitive matches between the violations returned by ECHO and the grower's entity name or address. If we find a match, the entity has failed the review, otherwise the entity has passed. More specifically,

- If the EPA ECHO web app returns warnings or violations issued to grower entity, they fail.
- If the EPA ECHO web app does *not* show warnings/ violations issued to grower entity **or** produces *no* results, they pass.

Regardless of whether or not there is a violation connected to the grower/ entity, a PDF file of the search results goes into our records as proof of confirmation. The following information is then stored in our centralized agronomic database for each manually reviewed grower: agronomy data service (ADS).

- Indication of pass or fail
- PDF of the EPA ECHO web app search results
- Summary of the nature of the violations (when applicable)

1.2.1 Level 1: Event establishment (L1)

Level 1 looks for the number of events present within a given calendar year for a given field. It is designed to help the Carbon Customer Success team understand how well data entry is going, determine which growers may benefit from assistance, gauge the amount of evidence review that will likely be needed, and most importantly ensure that the appropriate amount of historical data are entered to create the baseline.

This check looks primarily at historical data entry (i.e., in the baseline period). L1 is assessed on a "management zone" level and then aggregated to the field. (For details on management zones, please see Section 6.1 of the Monitoring Plan.) A minimum of 95% of the field must be covered by events to be considered complete.

Recall from the SEP v1.1 that growers must provide a minimum of three years of historical information, or a full crop/ management practice rotation, up to a maximum of five years. For each historical year, the grower must confirm *one* of the following for each practice.

- *Use* of a practice (i.e., add an event with attributes – a *positive* event),
- *Absence* of a practice (i.e., specify None/ Did Not Occur – a *negative* event)

NASS CDL information is used to determine the approximate number of crops in rotation and thus the approximate number of historical years expected. As noted in Gate 1, there are eight core agronomic event types, each with 3 – 5 years of historical data, and, therefore 21 (3 years), 27 (4 years), or 35 (5 years) total (positive + negative) events.

Table 6 shown below lists the attributes that are "required if practice changing" for all *positive* versions of each event type. If a given event is missing one or more of these "required" attributes, or if the event is missing entirely and cannot be gapfilled, the field will fail L1. (The term "required" here means that, if the event occurred, all management practice information must be submitted by the Field Manager; it is not considered supplemental.)

Table 6. Attributes for each event type that are considered required if practice changing for all growers and all fields in the Project. For more information on these events and attributes, see IndigoCarbon_US-1_2023_0029.

Event Type	**Required if Practice Changing Attributes**
Cash Crop Planting	• Crop type – What crop did you plant in this field? * • Planting date – When was this crop planted? **
Cover Crop Planting	• Cover crop species – What species did you plant in this field? * • Planting date – When was this crop planted? **
Tillage	• Tillage date – On what date did tillage occur? ** • Tillage method – What tillage method did you use? *
Chemical Applications (specifically Fertilizer)	• Product – What fertilizer was applied? * • Application rate – How much did you apply? ** • Application date – On what date did you apply this fertilizer? ** • Nitrification Inhibitor – Did you apply a nitrification inhibitor?* • Urease Inhibitor – Did you apply a urease inhibitor? * • Application method – What was your application method? *
Harvest or Termination/ Failure	• Crop type – What crop was harvested? * • Harvest/ termination date – When was the crop harvested/ terminated? ** • Yield – What was the crop yield? ** • Residue Burning – Was the crop residue on this field burned? *
Organic Amendments (including manure)	• Type – What type of organic amendment did you apply? * • Application date – On what date did you apply this organic amendment? ** • Application rate – How much did you apply? **

* Qualitative attribute, see Section 6.1 of SEP v1.1.

** Quantitative attribute, see Section 6.1 of SEP v1.1.

CDL crop history determination Since L1 needs to "know" a rotation ahead of grower data entry, we use CDL to determine a reasonable guess at the number of years of data to collect in the historical baseline period. Per the SEP v1.1 Section 3.4.1.3, "the length of the historical period shall be no less than three years, and shall at least be long enough to encompass a complete rotation of crops **and** management practices, unless a complete rotation extends beyond five years (e.g., if the same crop is grown every year, but the field is only tilled every four years, the historical period must be at least four years)." The table below explains how we determine management rotations for each project activity.

Table 7. Approach to determining management rotations for each project activity.

Project Activity	Approach
Crop planting and harvesting (automated check)	We do not define management practices based on this category because of guidance provided in Appendix E.8 of the SEP v1.1: For fields which were previously left fallow between major crop growing seasons, and in the project scenario are no longer fallow (e.g., planting winter cover crops), this practice change does not affect the rotation as it relates to the determination of the baseline.
Fertilizer management (automated check)	We account for fertilizer management by determining whether the crop followed a legume or not (as noted above).
Tillage (manual check) *	We determine a tillage rotation by assessing whether there was a tillage event associated with each crop.

* For manual check details, see the section below on "Agronomic Override."

Determining "distinct" crops The crop rotation algorithm is based on determining the number of "distinct" crops in a specific period, roughly 6 years prior to the field start date (e.g., 2015 for a 2021 field start). For each (calendar) year (of the harvested crop) between 2016 and the practice change year indicated by the grower (Y_0), we assess the "distinct" crops in the 5 preceding years (i.e., Y_0, Y_{-1}, Y_{-2}, Y_{-3}, Y_{-4}, Y_{-5} (Y_{-6})) where $Y_{-5} = 2017$ if $Y_0 = 2022$ or $Y_{-5} = 2015$ for $Y_0 = 2020$.

Here, a "distinct crop" is defined as the *cash* crop, and if the crop were a non-legume, whether it followed a legume or non-legume in the preceding year. More specifically, it's the difference between corn-following-corn (denoted "corn") and corn-following-soy (denoted "corn(L)"). For that reason, Y_{-6} will be necessary to determine if Y_{-5} should be denoted with "(L)."

This logic holds up under double cropping as well, as long as we consider the two crops in the double crop to be a single distinct cash crop. For example, wheat–soy–corn(L)–wheat (i.e., soy/corn planted in *different* years) would be different than wheat–soy/corn(L)–wheat (i.e., soy/corn planted in the *same* year).

Recall that for this L1 check, we are *not* looking at Y_1 or Y_2 (i.e., years *post*-practice change) because (1) one additional practice is adding a new crop to the rotation, and (2) the order of the baseline crop rotation determines whether we construct a matched baseline or a blended baseline. (For information on baselines, see IndigoCarbon_US-1_2023_0048.)

Connecting distinct crops to historical data needs The required number of historic years depends on the order and the number of distinct crops as outlined below.

- If **5 distinct crops** are identified, 5 years of historic information are required.
- If **4 distinct crops** are identified,
  - And appear in Y_{-1} through Y_{-4} **and** if Y_{-1} is the same as Y_{-5} (providing a repeating rotation is detected), 4 years of historic information are required.
  - Otherwise, 5 years of historic information are required.
- If **3 distinct crops** are identified,
  - And appear in Y_{-1} through Y_{-3} **and** Y_{-1} is the same as Y_{-4} **and** Y_{-2} is the same as Y_{-5} (providing evidence of a repeating crop rotation), 3 years of historic information are required.
  - And appear in Y_{-1} through Y_{-4} **and** if Y_{-1} is the same as Y_{-5} (providing some evidence of a repeating rotation), 4 years of historic information are required.
  - Otherwise, 5 years of historic information are required.
- If **2 distinct crops** are identified,
  - And if Y_{-1} is the same as Y_{-3} is the same as Y_{-4} **and** Y_{-2} is the same as Y_{-5} (providing some evidence of a repeating rotation), 3 years of historic information are required.

  - And if the 5 years could conceivably be a repeating sequence of 4 or a repeating sequence of 2, 4 years of historic information are required.
  - Otherwise, 5 years of historic information are required.
- If **1 distinct crop** is identified, 3 years of historic information are required.

Management practices required to determine distinct crop rotations For simplicity, the above examples are all provided for rotations where a grower tilled each year. In the strictest sense, the historical period is determined by *both* the crop rotation *and* the management rotation. In other words, "crop type" and "management" fold together in the concept of a "pattern of crop cultivation activity."

Defining management practice periodicity for the purposes of defining required historic information only matters if the practice change that is being credited requires a certain specific period of historic information to provide evidence that the change occurred, i.e., tilling. Therefore we must also use information about whether tillage was used before each crop to identify these "distinct" crops. Specifically for tillage, whether the crop was in a no-till (NT) year or a year with tillage. Tillage events associated with a crop are those occurring after harvest of the previous crop and before harvest of the current crop.

Agronomic override of required history

Indeterminant crop history We use CDL to determine how many years of historical data should be expected from the grower. However, in some instances, CDL is unable to determine exactly *what* the historical crop rotation has been. In those instances, we default to asking the grower for the maximum 5 years of historical data, while in fact they may only be required to submit 3 or 4 years. To address this potential "over-ask," CCS manually investigates such fields to see if, based on what the grower has submitted thus far, they can determine internally what their crop rotation is. If we can determine that it is a single-crop (e.g., corn-corn-corn) or a 2-crop rotation (e.g., corn-soybeans-corn-soybeans), they can override and reduce the required years. The following standard operating procedure is employed to assess if (and when) such an override is necessary.

CCS checks the pattern of planting events, starting with the year before the indicated practice change. If in reviewing the entered crop rotation and based on the guiding logic noted below, CCS decides to override the 5 years of required history with fewer years.

- If they see a grower has planted a single crop for at least 3 consecutive years, override to 3 years.
- If they see a grower has had a 2-crop rotation for 2 rotations, override to 4 years.

Ideally, we want to analyze the crop rotation starting with the year before the practice change because diversifying crop rotation is a practice change. If the grower has not entered adequate historical years but they're able to determine the crop rotation based on post-practice change data entry, it is also permissible to use that as grounds for an override.

Record of this override and the rationale of each override is recorded in our centralized agronomic database for each manually reviewed field (agronomy data service, ADS). Specifically that includes (1) what crop rotation they observed over how many years, (2) who was consulted (if an agronomic specialist assisted), and (3) any other information used to determine crop rotation.

Tillage management practices We use CDL to automatically determine how many years of history to ask from the grower; however, it does not provide management data like tillage events. As a result, a grower's field may initially require 3 or 4 years of agronomic history, based on the CDL-determined rotation alone, but may actually require an additional year (or 2), based on their practice rotation (e.g., no-till events). We use manual review as a way to determine the full "pattern of crop cultivation activity." Similar to the indeterminant situation noted above, we can override and *increase* the required years. The following standard operating procedure is employed to assess if (and when) such an override is necessary.

Much like before, CCS will check for the year that the practice change was indicated, the current number of historical years required based on CDL, and the (management) rotation that the grower has provided up to the year of the practice change. They will assign "distinct" crops as noted above, accounting for legumes (L) and no-till (NT) activities. Based on the required history associated with these district crops as noted above, CCS will increase the historical years required to 4 or 5 years.

Record of this override and the rationale of each override is recorded in our centralized agronomic database for each manually reviewed field (agronomy data service, ADS). Specifically that includes (1) what crop/ practice rotation they observed over how many years, (2) who was consulted (if an agronomic specialist assisted), and (3) any other information used to determine crop/ practice rotation.

1.2.2 Level 2: Conflicting logic

Once the events are in place, we can compare among event types to determine if any one event type is in conflict with another. There are several impossibilities for which we can check. As with L1, L2 is assessed on a "management zone" level and then aggregated to the field. (For details on management zones, please see Section 6.1 of the Monitoring Plan). A minimum of 95% of the field must be covered by events to be considered complete.

Geometry errors

- Geometries within a given event type sum to (95% of the) field boundary.
- Geometries for (cash crop) planting events do not overlap in space and time simultaneously (excluding replants).

Duplicate event errors For any and all event types, the same event type cannot have both positive and negative events for a given agronomic year (i.e., a grower cannot claim that they simultaneously *did do* and *did not do* the same thing on the same part of the field at the same time).

Date mismatch errors

- Planting date must be *before* harvest or termination/ failure date for same agronomic year.
- Harvest or termination/ failure date must be *after* planting date for same agronomic year.
- For irrigation events, the start date must occur *before* the end date for same agronomic year.
- For irrigation events, the end date must occur *after* the start date for same agronomic year.

Crop type mismatch errors The cover crop species must match the planted cover crop for same agronomic year.

Invalid method errors Only tillage methods associated with (very) low impact categories are allowed between planting and harvest dates. (For a list of which tillage methods are associated with which impact categories, see IndigoCarbon_US-1_2023_0029.)

Missing companion event errors If the termination method is "tillage," there must be a companion tillage event present that is dated within 1d of the termination.

If a fertilizer or manure application method is "incorporated" (e.g., 0–4 in, 5–12 in, and 13+ in), there must be a companion tillage event present that is dated within 1d of the application and have a method that is consistent with the incorporation depth.

- The application method cannot be incorporated if the tillage method is associated with the very low impact category.
- For incorporations of 0–4 in, tillage methods associated with the low and medium impact categories are allowed.
- For incorporations of 5–12 in, tillage methods associated with the low, medium, and high impact categories are allowed.
- For incorporations of 13+ in, tillage methods associated with deep ripping or a high impact category is allowed.

(For a list of which tillage methods are associated with which impact categories, see IndigoCarbon_US-1_2023_0029).

1.2.3 Level 3: Practice change assessment (L3)

As noted in SEP v1.1, the project area mandates additional or different data requirements, depending on the management situation. Practice change activities require more information than do the "basic" versions of each event type for a given management practice. In other words, information beyond what is noted for L1 is required for L3. As with L1 and L2, L3 is assessed on a "management zone" level and then aggregated to the field. (For details on management zones, please see Section 6.1 of the Monitoring Plan). Up to a total of 9 additional attributes (per year) become required (i.e., not supplemental because indicated as practice changing) based on the practice change indicated by the grower and are detailed in Table 2. For some practice changes, we need the event details *before* the declared change date (e.g., tillage to prove a reduction is possible). For others, we need the required event information *after* the declared change date (e.g., cover crop planting to show an increase in cover crops planted is possible).

Table 2. Attributes required to assess practice change activities.

Practice Change	**Applicable Event Type**	**Additional Event Attributes**
Inorganic nitrogen fertilizer application	Chemical Applications (i.e., fertilizers)	• Application rate • Application method
Soil disturbance and/or residue management	Tillage	Tillage method
Cropping practices	Cash Crop Planting, Cover Crop Planting, Harvest, and Termination/ Failure	None – Required attributes are sufficient

1.2.4 Level 4: Data confidence (L4)

Once the field-level data have passed data completeness and logic checks (Levels 1 – 3), they can move to Level 4, which is where we review the actual attribute values and selections added by the grower through the Carbon by Indigo user interface.

Turnstiles Within this level, there is also a series of turnstiles through which the data must pass, referenced in Section 6.7 of the Monitoring Plan (Figure 6.7) and Section 6.1 of the SEP v1.1. Here we review the data values and compare them against remote sensing algorithm results and other geospatial datasets (Turnstile 2), third-party data sources (Turnstile 3), ranges determined by subject matter experts (Turnstile 4), ranges determined to be agronomically feasible by subject matter experts (Turnstile 5), and the presence (or absence) of exhibits (i.e., evidence/ supporting documentation in Gate 3). To pass L4, the individual event attribute has to pass at least *one* turnstile among T2, T3, or T4 and pass T5. For the situations where the source data are missing and thus the check cannot be performed, we default to other turnstiles and/or similar checks from L2. (A summary of these turnstile checks and their relationships to each other and L2 is included at the end of this section.)

Turnstile 2: Remote sensing data (L4T2) Turnstile 2 references database tables generated by field-scale data products used to evaluate grower data management events against model predictions provided by remote sensing algorithms. Data products were developed by Indigo's Remote Sensing Group and include: (1) cover crop detection, (2) tillage detection, (3) irrigation detection, and (4) harvest date. For all of the detection checks (1-3), we confirm the presence and/or absence of an event, and if present, we confirm the date range. In other words, remote sensing provides a true/ false output (i.e., detected or not detected) for a given time period, usually by season or year.

There is also a "remote sensing" check for crop type. (As discussed more below, we also check against crop type because it is fundamental to our project definition.) The remote sensing algorithms and methodology used in developing the data products are described in IndigoCarbon_US-1_2023_0055. For more information on CDL, see IndigoCarbon_US-1_2023_0069.

Remote sensing model predictions are specific to the Crop Management Zone (CMZ), and predictions were only used if the F1 ("fitness") score was greater than or equal to 0.7. (See IndigoCarbon_US-1_2023_0055.) If the model prediction agreed with the grower-entered data, the field passed Turnstile 2. Where the model prediction was in conflict with the grower entered data, the field was required to pass one of the other checks at T3 or T4 to move on to Level 5. (See Table 18 for Turnstile comparisons.)

In cases where no model prediction was available (due to a low F1 score or no coverage for that CMZ), or where the model prediction was in conflict with the grower data and there were no other Level 4 check, we relied on the Field

Manager's attestation and evidence provided to ensure that the data they entered were correct. Negative events (e.g., grower said tillage did not happen) are considered qualitative data and did not have secondary quality checks in Level 4; all quantitative data (e.g., cover crop planting date) were subjected to a secondary check in Level 4.

Turnstile 3: 3rd-party data sources (L4T3) Turnstile 3 conforms to the data sources described in SEP v1.1 data and evidence hierarchy (2): agricultural census data and published literature. It references NASS, Agricultural Resource Management Survey (ARMS), or other relevant, peer-reviewed third-party data sources to establish ranges used to check grower data. The grower data values must fall within these ranges to pass the check. The types of grower data relying on 3rd-party data sources include nitrogen fertilizer rate, percentages of nutrients in fertilizer blends, yield, planting and harvest dates, and organic amendment rate. The ranges and methods used in deriving the ranges are provided below for each type of grower data.

Nitrogen fertilizer rate

3rd-party data sources used

1. NASS Quick Stats: Further information about NASS Quick Stats can be found in IndigoCarbon_US-1_2023_0069.
2. Nitrogen Management Guidelines for Corn in Indiana (Purdue University), available at http://www.kingcorn.org/news/timeless/NitrogenMgmt.pdf.
3. Nutrient Recommendations from Michigan State University, available at https://www.canr.msu.edu/fertrec/uploads/E-2904-MSU-Nutrient-recomdns-field-crops.pdf.

N Fertilizer Rate Method: Mean N fertilizer rate by crop, state, and sometimes ARMS region (see IndigoCarbon_US-1_2023_0069 for description) was obtained from NASS Quick Stats. These values served as the means of the N rate ranges used for each crop–region pair in the Project. A coefficient of variation (CV) for optimal N rates on corn was first approximated using an N rate calculator published by Purdue (link #2 above). Regional estimates for minimum and maximum optimal N rate were obtained using Tables 2–7 in the Purdue document, which provided economically optimal N rates for corn in Indiana based on a range of corn and nitrogen prices. These minimum and maximum values of optimal N rate were then used to represent the 68% interval of actual N rates, with the standard deviation assumed to be half the width of this interval. The CV for corn based on the Purdue tables was 44.1%.

This general method of determining a CV based on recommended N rate was also applied to nutrient recommendations published by Michigan State University (link #3 above), with a resulting CV value of 53.9% for corn. The Purdue value was chosen over the MSU value because it was smaller, thereby establishing a narrower range to use in the check of N fertilizer rate. The CV value of 44.1% was then applied to all crops using the mean NASS N rate for that crop per ARMS region, when region was available (Geographic Unit = "region" in Table 3 below). When region was not available, the CV was applied to the mean for that crop across all states included in the Project (Geographic Unit = "project" in Table 3). Finally, a 99% interval was used to obtain the final range for each crop–region pair. All low-end range values were zero, and upper-end range values reflective of typical N rates for that region.

Table 3. Ranges used for N fertilizer rate (in lbs/ac) by region.

ARMS Region	Crop	Mean	SD	99% Lower Interval	99% Upper Interval	Geographic Unit
Atlantic	BARLEY	53	23	0	112	region
Midwest	BARLEY	55	24	0	118	region
Plains	BARLEY	65	29	0	140	region
South	BARLEY	64	28	0	138	project
West	BARLEY	119	52	0	254	region
Atlantic	CORN	116	51	0	249	region
Midwest	CORN	133	58	0	283	region
Plains	CORN	129	57	0	276	region
South	CORN	144	64	0	308	region
West	CORN	132	58	0	282	region
Atlantic	COTTON	86	38	0	184	region
Midwest	COTTON	106	47	0	226	region
Plains	COTTON	71	31	0	152	region
South	COTTON	99	44	0	212	region
West	COTTON	93	41	0	199	project
Atlantic	OATS	30	13	0	65	region
Midwest	OATS	36	16	0	76	region
Plains	OATS	56	25	0	119	region
South	OATS	43	19	0	91	project
West	OATS	43	19	0	91	project
Atlantic	PEANUTS	23	10	0	48	region
Midwest	PEANUTS	35	15	0	74	project
Plains	PEANUTS	64	28	0	138	region
South	PEANUTS	24	10	0	50	region
West	PEANUTS	35	15	0	74	project
Atlantic	SORGHUM	71	31	0	152	project
Midwest	SORGHUM	117	52	0	250	region
Plains	SORGHUM	73	32	0	157	region
South	SORGHUM	71	31	0	152	project
West	SORGHUM	43	19	0	93	region
Atlantic	SOYBEANS	27	12	0	58	region
Midwest	SOYBEANS	21	9	0	45	region
Plains	SOYBEANS	19	9	0	41	region
South	SOYBEANS	26	12	0	56	region
West	SOYBEANS	23	10	0	49	project
Atlantic	SUGARBEETS	102	45	0	217	project
Midwest	SUGARBEETS	110	48	0	234	region
Plains	SUGARBEETS	100	44	0	214	region
South	SUGARBEETS	102	45	0	217	project
West	SUGARBEETS	90	40	0	192	region
Atlantic	SUNFLOWER	63	28	0	135	project
Midwest	SUNFLOWER	63	28	0	135	project
Plains	SUNFLOWER	63	28	0	135	region
South	SUNFLOWER	63	28	0	135	project
West	SUNFLOWER	63	28	0	135	project
Atlantic	WHEAT	89	39	0	190	region
Midwest	WHEAT	93	41	0	198	region
Plains	WHEAT	65	29	0	138	region
South	WHEAT	105	46	0	224	region
West	WHEAT	41	18	0	88	region

Yield

3rd-party data sources

1. NASS Quick Stats: Further information about NASS Quick Stats can be found in IndigoCarbon_US-1_2023_0069.
2. Tack, Jesse; Coble, Keith H.; Johansson, Robert; Harri, Ardian; and Barnett, Barry, The Potential Implications of "Big Ag Data" for USDA Forecasts (January 31, 2017). Available at SSRN: https://ssrn.com/abstract=2909215 or http://dx.doi.org/10.2139/ssrn.2909215.

Yield Method: Similarly to the approach used to derive N rate ranges, an estimate of the distribution of field-level yield values was applied to average yield estimates from NASS Quick Stats (see IndigoCarbon_US-1_2023_0069 for description) to produce confidence intervals around those means and thus derive appropriate ranges to be used in checking yield. A CV was derived from Tack et al. (2017), specifically Table 1 that provided corn yield, acreage, and production data for over 156,000 U.S. farms in 47 states during a 10-year period from 1999–2008. The CV obtained from this dataset was 0.30 and was used to derive a 99% confidence interval that was then applied to the average state-level NASS yield for all crops and all states in the Project. Tables 4 and 5 below provide ranges for two example crops (corn and soybean); however, the same method was used to derive ranges for all crops in use in the Project.

Table 4. Ranges used for corn yield (in bu/ac) by state.

State	Average Yield	Minimum Yield	Maximum Yield
AL	140.8	30.4	251.3
AR	175	37.7	312.3
AZ	203.9	44	363.9
CA	178.4	38.5	318.4
CO	135.1	29.1	241
DE	161.4	34.8	288
FL	136.3	29.4	243.2
GA	169.8	36.6	303
IA	182.1	39.3	324.9
ID	198.7	42.8	354.5
IL	179.9	38.8	321
IN	167	36	298
KS	129.8	28	231.7
KY	157.1	33.9	280.3
LA	168.7	36.4	301
MD	149.3	32.2	266.5
MI	154.7	33.3	276
MN	176.6	38.1	315.1
MO	144.6	31.2	258
MS	170.3	36.7	303.9
MT	104.5	22.5	186.6
NC	119.4	25.7	213.1
ND	129.1	27.8	230.3
NE	175.4	37.8	313
NJ	142	30.6	253.4
NM	172.4	37.2	307.6
NY	148.2	31.9	264.4
OH	165.2	35.6	294.7
OK	129.9	28	231.8
OR	209.2	45.1	373.3
PA	142.3	30.7	253.8
SC	114.3	24.7	204
SD	143.4	30.9	255.9
TN	152.1	32.8	271.4
TX	129.9	28	231.8

State	Average Yield	Minimum Yield	Maximum Yield
UT	167	36	298
VA	133.6	28.8	238.4
WA	221.7	47.8	395.5
WI	161.4	34.8	288
WV	139.5	30.1	248.8
WY	139.1	30	248.2

Table 5. Ranges used for soybean yield (in bu/ac) by state.

State	Average Yield	Minimum Yield	Maximum Yield
AL	39	8.4	69.7
AR	46.5	10	83
DE	43.6	9.4	77.8
FL	36.1	7.8	64.4
GA	36.4	7.8	64.9
IA	53.5	11.5	95.5
IL	55.2	11.9	98.5
IN	52.6	11.3	93.8
KS	37.2	8	66.4
KY	47.5	10.3	84.8
LA	48.1	10.4	85.9
MD	44.1	9.5	78.7
MI	45.5	9.8	81.2
MN	46	9.9	82.2
MO	43.4	9.4	77.5
MS	48.4	10.4	86.3
NC	35.2	7.6	62.8
ND	33.1	7.1	59.1
NE	55.7	12	99.4
NJ	38.6	8.3	68.8
NY	46.9	10.1	83.6
OH	51.1	11	91.2
OK	26	5.6	46.5
PA	46.6	10.1	83.2
SC	30.6	6.6	54.6
SD	42	9.1	74.9
TN	43.9	9.5	78.4
TX	30	6.5	53.6
VA	38.5	8.3	68.8
WI	48	10.3	85.6
WV	47.3	10.2	84.4

Planting and harvest date

NASS Usual Planting and Harvest Dates for U.S. Field Crops (2010): Further information about the NASS Usual Planting and Harvest Dates handbook can be found in IndigoCarbon_US-1_2023_0069.

Planting and Harvest Date Method: The "Begin" and "End" dates provided in the NASS handbook for both planting and harvest were used to establish the (usual) ranges used in checking grower-provided dates for a given crop–state pair.

From the NASS usual planting/ harvest dates, we pull the planting start date (i.e., when 5% of planting has begun) and the harvest start date (i.e., when 95% of harvest is complete). If data are not available for the given crop–state pair, then we may check a proxy state. (A proxy state is a nearby state having similar geography, climate, and/or agricultural practices. For example, the proxy states for Illinois are Indiana, Iowa, and Missouri.)

If the above fails (i.e., there is no estimated range or the grower data fall outside of the "usual" range), then we look for a "best date" estimate. (The best date estimate is where 50% of planting/ harvest is complete.) If a best date is found, then we construct a ± 90d range around this best date, and check if the grower data fall within this range.

% Nitrogen (N), phosphorus (P_2O_5), and potassium (K_2O)

Chemical Data Management System (CDMS) database: Details of the database and its implementation can be found in IndigoCarbon_US-1_2023_0069.

%NPK Method: The ranges for checking percentages of N, P (i.e., P_2O_5), K (i.e., K_2O), and other micronutrients were derived using the CDMS database. (See IndigoCarbon_US-1_2023_0069 for description.) The ranges for each nutrient were defined as the minimum and maximum values found for each nutrient in the database, respectively. The minimums were all zero because every component in a custom fertilizer blend had at least one product where it was not present.

Table 6. Ranges used for nutrient percentages.

Plant Nutrient	No. Products in CMDS	Avg.	Std.	Min.	Max.
Total Nitrogen (N)	973	1.5%	4.9%	0.0%	46.0%
Available Phosphate (P_2O_5)	450	0.9%	4.8%	0.0%	60.0%
Soluble Potash (K_2O)	616	0.9%	4.3%	0.0%	53.0%
Sulfur (S)	488	0.5%	3.5%	0.0%	99.9%
Zinc (Zn)	562	0.3%	2.3%	0.0%	52.0%
Calcium (Ca)	207	0.3%	2.1%	0.0%	35.0%
Magnesium (Mg)	198	0.1%	0.8%	0.0%	24.0%
Iron (Fe)	428	0.1%	1.1%	0.0%	50.0%
Manganese (Mn)	445	0.1%	1.2%	0.0%	35.0%
Copper (Cu)	274	0.1%	0.7%	0.0%	33.0%

Organic amendment rate

Agricultural Resource Management Survey (ARMS): Further details about the ARMS survey can be found in IndigoCarbon_US-1_2023_0069.

Organic Amendment Rate Method: Manure application rate data from ARMS (see IndigoCarbon_US-1_2023_0069 for description) was used to establish a single range used in checking grower-provided rates of organic amendments. While regional and crop-specific data were provided in the ARMS survey, there were relatively few survey respondents representing each region and crop pair. Thus, we chose to aggregate the survey results across all regions and crops to obtain a single minimum and maximum value to define the range. The minimum and maximum rates represent amounts of manure applied, either in dry weight (tons) or liquid volume (gallons).

Table 7. Range used for organic amendment rate per acre.

Minimum Rate (ton)	Maximum Rate (ton)	Minimum Rate (gal)	Maximum Rate (gal)
0	48.5	0	12128

Turnstile 4: SME-derived agronomic ranges (L4T4) Indigo is a science-based company founded on scientific principles, following best practices to ensure rigor, with extensive in-house scientific expertise. As such we have a number of subject-matter experts (SMEs) in the fields of agronomy, agronomic science, and farm management. (See SME Credentials below.) From the decades of collective personal and professional experience, they have derived a series of crop-specific agronomic ranges and acceptable selections with which to further assess the quality of grower-entered responses and are given the following tables. In addition to these ranges, they also provided several "rules of thumb" applicable to all crops.

SME ranges general to all crops

Irrigation Summary (seasonal occurrence) The start date cannot be earlier than 4 weeks before the planting date, and the end date cannot be later than the harvest date. Irrigation is considered a seasonal practice, and thus the irrigation event includes a date range during which irrigation was applied. The date range can overlap with many weeks, during which individual irrigation events were assumed to occur.

Chemical Applications (e.g., fertilizers)

Fertilizer application cannot occur earlier than harvesting of the previous crop and cannot be later than 4 weeks prior to harvesting the "current" crop, unless it is silage corn, then fertilization up through harvest is allowed.

Dry fertilizers cannot be applied at rates exceeding 1000 lbs/ ac. Liquid fertilizers cannot be applied at rates exceeding 100 gal/ ac. The "anhydrous" fertilizer cannot be applied at rates in excess of 300 lbs/ ac.

For growers who enter custom product blends, the "N-P-K" component requirements are as follows. If a grower indicates a product (i.e., commercial, generic, or custom) with ammonium nitrate as their form of nitrogen, it is broken down into 50:50 ammonium and nitrate.

- Nitrogen
  - %N $<$ 46% if dry
  - %N $<$ 32% if wet
  - %N $<$ 82% if anhydrous
- $\%P_2O_5 < 61\%$
- $\%K_2O < 62\%$
- $\%N + \%P_2O_5 + \%K_2O < 100\%$

Organic Amendments (e.g., manure)

Application cannot occur earlier than harvesting of the previous crop and cannot be later than 4 weeks prior to harvesting the "current" crop, unless it is silage corn or winter wheat, then fertilization up through harvest is allowed.

Liquid (or slurry) manures cannot be applied at rates exceeding 25K gal/ ac, and dry (solid) manures cannot exceed application rates of 25 ton/ ac.

Grazing

Grazing is defined when the crop is NOT growing (i.e., between previous harvest and planting). In the scenario where wheat is planted and harvested before April 1, all grazing dates are considered valid. Additional attribute ranges are displayed in the table.

Table 8. SME ranges for grazing attributes [units].

Event Attribute	**Range**
Stocking count *	0.1 / 8 [head/ac]
Stocking weight *	100 / 10,000 [lbs/ac]
Supplemental feed rate **	0.125 / 30 [lbs/ac/standard head/day]
Grazing days	0 / 146 [days/ac/yr]
Grazing hours per day	0 / 24 [hrs/day]

* Growers must provide either stocking count or stocking weight.

** Supplemental feed rate is optional. Assume that a standard head is equivalent to 1,000 lbs.

Harvest or Termination/ Failure

For harvest events, the harvest yield must be greater than zero, otherwise it should be a termination/ failure event. Cover crop termination is allowed up to 14d after planting of the next cash crop.

Crops that are harvested for silage, haylage, or hay (so-called "forage-harvest crops") tend to have much larger yields than those harvested for grain-only because more of the plant is harvested. (See

Fig. 1 from M. Lippert (2016) to compare the differences between silage corn yield and grain corn yield ranges.) In reality most of the management practices between grain-only and forage-harvest crops are the same; it's really just the yields that differ significantly. For this reason, we generated yield ranges that are general to all crops that have harvest methods of silage, haylage, or hay.

Table 9. SME yield ranges for any forage-harvest crop. All ranges are in terms of minimum/ maximum with associated [units] at a percentage [%] of dry matter.

Harvest Method	**Harvest Yield**
Silage	0 / 35 [tons/ac @ 35%]
Haylage	0 / 15 [tons/ac @ 35%]
Hay	0 / 10 [tons/ac @ 85%]

1.2.5 SME ranges for specific crops

Table 10. SME ranges for corn, soybeans, cotton, wheat, triticale, and peanuts. All ranges are in terms of min/ max with associated [units].

Event Attribute	**Corn**	**Soybeans**	**Cotton**	**Wheat**	**Triticale**	**Peanuts**
Harvest yield (grain)	0 / 400 [bu/ac]	0 / 140 [bu/ac]	0 / 2200 [lbs/ac]	0 / 140 [bu/ac]	0 / 140 [bu/ac]	0 / 7000 [lbs/ac]
Harvest yield (hay)	0 / 10 [ton/ac]	0 / 10 [ton/ac]	0 / 10 [ton/ac]	0 / 10 [ton/ac]	0 / 10 [ton/ac]	0 / 10 [ton/ac]
Harvest yield (haylage)	0 / 15 [ton/ac]	0 / 15 [ton/ac]	0 / 15 [ton/ac]	0 / 15 [ton/ac]	0 / 15 [ton/ac]	0 / 15 [ton/ac]
Harvest yield (silage)	0 / 35 [ton/ac]	0 / 35 [ton/ac]	0 / 35 [ton/ac]	0 / 35 [ton/ac]	0 / 35 [ton/ac]	0 / 35 [ton/ac]
Tillage depth	0 / 18 [in]	0 / 18 [in]	0 / 18 [in]	0 / 18 [in]	0 / 18 [in]	0 / 18 [in]
% N in product per event blend component	0 / 82 [%]	0 / 46 [%]	0 / 82 [%]	0 / 82 [%]	0 / 82 [%]	0 / 46 [%]
Total N rate per crop growing season	30 / 350 [lbs/ac]	0 / 50 [lbs/ac]	0 / 200 [lbs/ac]	0 / 175 [lbs/ac]	0 / 150 [lbs/ac]	0 / 150 [lbs/ac]
% P_2O_5 in product per event blend component	0 / 61 [%]	0 / 61 [%]	0 / 61 [%]	0 / 61 [%]	0 / 61 [%]	0 / 61 [%]
Total P_2O_5 rate per crop growing season	0 / 200 [lbs/ac]	0 / 200 [lbs/ac]	0 / 200 [lbs/ac]	0 / 200 [lbs/ac]	0 / 150 [lbs/ac]	0 / 150 [lbs/ac]
% K_2O in product per event blend component	0 / 62 [%]	0 / 62 [%]	0 / 62 [%]	0 / 62 [%]	0 / 62 [%]	0 / 62 [%]
Total K_2O rate per crop growing season	0 / 200 [lbs/ac]	0 / 200 [lbs/ac]	0 / 200 [lbs/ac]	0 / 200 [lbs/ac]	0 / 180 [lbs/ac]	0 / 180 [lbs/ac]

Event Attribute	**Corn**	**Soybeans**	**Cotton**	**Wheat**	**Triticale**	**Peanuts**
% S in product per event blend component	0 / 90 [%]	0 / 90 [%]	0 / 90 [%]	0 / 90 [%]	0 / 90 [%]	0 / 90 [%]

Table 11. SME ranges for alfalfa, beans/ peas, sorghum, sunflowers, canola, and millet. All ranges are in terms of min/ max with associated [units].

Event Attribute	Alfalfa	Beans/ Peas	Sorghum	Sunflowers	Canola	Millet
Harvest yield (grain)	0 / 10 [ton/ac]	0 / 8400 [lbs/ac]	0 / 180 [bu/ac]	0 / 2000 [lbs/ac]	0 / 3600 [lbs/ac]	0 / 140 [bu/ac]
Harvest yield (hay)	0 / 10 [ton/ac]	0 / 10 [ton/ac]	0 / 10 [ton/ac]	0 / 10 [ton/ac]	0 / 10 [ton/ac]	0 / 10 [ton/ac]
Harvest yield (haylage)	0 / 15 [ton/ac]	0 / 15 [ton/ac]	0 / 15 [ton/ac]	0 / 15 [ton/ac]	0 / 15 [ton/ac]	0 / 15 [ton/ac]
Harvest yield (silage)	0 / 35 [ton/ac]	0 / 35 [ton/ac]	0 / 35 [ton/ac]	0 / 35 [ton/ac]	0 / 35 [ton/ac]	0 / 35 [ton/ac]
Tillage depth	0 / 18 [in]	0 / 18 [in]	0 / 18 [in]	0 / 18 [in]	0 / 18 [in]	0 / 18 [in]
% N in product per event blend component	0 / 46 [%]	0 / 82 [%]	0 / 82 [%]	0 / 82 [%]	0 / 82 [%]	0 / 82 [%]
Total N rate per crop growing season	0 / 150 [lbs/ac]	0 / 150 [lbs/ac]	0 / 200 [lb/ac]	30 / 150 [lbs/ac]	0 / 170 [lbs/ac]	0 / 150 [lbs/ac]
% P_2O_5 in product per event blend component	0 / 61 [%]	0 / 61 [%]	0 / 61 [%]	0 / 61 [%]	0 / 61 [%]	0 / 61 [%]
Total P_2O_5 rate per crop growing season	0 / 150 [lbs/ac]	0 / 150 [lbs/ac]	0 / 150 [lbs/ac]	0 / 150 [lbs/ac]	0 / 150 [lbs/ac]	0 / 150 [lbs/ac]
% K_2O in product per event blend component	0 / 62 [%]	0 / 62 [%]	0 / 62 [%]	0 / 62 [%]	0 / 62 [%]	0 / 62 [%]
Total K_2O rate per crop growing season	0 / 200 [lbs/ac]	0 / 180 [lbs/ac]	0 / 180 [lbs/ac]	0 / 180 [lbs/ac]	0 / 180 [lbs/ac]	0 / 180 [lbs/ac]
% S in product per event blend component	0 / 90 [%]	0 / 90 [%]	0 / 90 [%]	0 / 90 [%]	0 / 90 [%]	0 / 90 [%]

Table 12. SME ranges for sugar beets, rye, oats, barley, and flax. All ranges are in terms of min/ max with associated [units].

Event Attribute	Sugar Beets	Rye	Oats	Barley	Flax
Harvest yield (grain)	0 / 40 [tons/ac]	0 / 140 [bu/ac]	0 / 160 [bu/ac]	0 / 200 [bu/ac]	0 / 55 [bu/ac]
Harvest yield (hay)	0 / 10 [ton/ac]	0 / 10 [ton/ac]	0 / 10 [ton/ac]	0 / 10 [ton/ac]	0 / 10 [ton/ac]
Harvest yield (haylage)	0 / 15 [ton/ac]	0 / 15 [ton/ac]	0 / 15 [ton/ac]	0 / 15 [ton/ac]	0 / 15 [ton/ac]

Event Attribute	Sugar Beets	Rye	Oats	Barley	Flax
Harvest yield (silage)	0 / 35 [ton/ac]	0 / 35 [ton/ac]	0 / 35 [ton/ac]	0 / 35 [ton/ac]	0 / 35 [ton/ac]
Tillage depth	0 / 18 [in]	0 / 18 [in]	0 / 18 [in]	0 / 18 [in]	0 / 18 [in]
% N in product per event blend component	0 / 82 [%]	0 / 82 [%]	0 / 82 [%]	0 / 82 [%]	0 / 82 [%]
Total N rate per crop growing season	30 / 350 [lbs/ac]	0 / 150 [lbs/ac]	0 / 150 [lbs/ac]	0 / 150 [lbs/ac]	0 / 100 [lbs/ac]
% P_2O_5 in product per event blend component	0 / 61 [%]	0 / 61 [%]	0 / 61 [%]	0 / 61 [%]	0 / 61 [%]
Total P_2O_5 rate per crop growing season	0 / 200 [lbs/ac]	0 / 150 [lbs/ac]	0 / 150 [lbs/ac]	0 / 150 [lbs/ac]	0 / 150 [lbs/ac]
% K_2O in product per event blend component	0 / 62 [%]	0 / 62 [%]	0 / 62 [%]	0 / 62 [%]	0 / 62 [%]
Total K_2O rate per crop growing season	0 / 310 [lbs/ac]	0 / 180 [lbs/ac]	0 / 180 [lbs/ac]	0 / 180 [lbs/ac]	0 / 180 [lbs/ac]
% S in product per event blend component	0 / 90 [%]	0 / 90 [%]	0 / 90 [%]	0 / 90 [%]	0 / 90 [%]

1.2.6 SME Credentials

The following Agronomy SMEs (listed alphabetically by last name) led to the development of the general and crop-specific SME ranges used in Turnstile 4.

Brian Bartle has over a decade of experience in the agriculture industry with a M.S. in Agriculture, Resource, and Environmental Economics from Michigan State University. Born and raised on a farm in Michigan, Bartle has experience in the public and private, as well as academic sectors, of agriculture, both domestically and internationally. His expertise includes farm financial management, farm data management, specialty crop production, organic agriculture, agriculture technology, and grain marketing, as well as general knowledge and application of soil science and crop sciences within the Great Lakes region.

Thomas Lawler has 10 years of professional experience working in the agriculture industry, with 3 years in the private industry, and the remainder working for Iowa State University. Lawler holds an M.S. in Soil Science and a B.S. in Agronomy from Iowa State University. His areas of expertise include soil fertility, soil biology, soil management, cover cropping, holistic agronomic planning, ag technology, ag education, and farm operational knowledge in the Midwest and Corn Belt. Lawler is also an active decision-maker on his family's 1,500-acre corn/soy farm in northwestern Iowa.

Daniel Ochs has 16 years of experience in the agriculture industry and has been a Certified Crop Advisor (CCA) since 2012. Ochs holds a B.S. in Applied Economics and Management from Cornell University and an M.S. in Ag, Food, and Resource Economics from Michigan State University. Ochs held positions with Scotiabank, New York Corn and Soybean Growers Association, Michigan State University Extension, and his family's crop consulting and seed sales agency before joining Indigo. His areas of expertise include farm business management, precision agriculture, and crop production of corn, soy, wheat, alfalfa, hay, and several specialty crops.

Darrin Unruh has 30 years of experience in the agriculture industry. Unruh grew up on a farm in south-central Kansas and took his first "Holistic Resource Management" course in the early 1990s. Since that time, Unruh has continuously run a small (less than 1,000-acre) family farm and ranch focused on biological farming and biomimicry. Unruh has experience in retail ag, including seed sales and feed sales; at one point he owned his own custom application business. Unruh is preeminent speaker in the agricultural community and has presented at many conferences on the topic of soil health, including Ohio No-Till Conference; ACRES USA Conference in Minneapolis, Minnesota; Soil Revolution Conference in Boulder, Colorado; and twice at Soil Health U Conference in Salina, Kansas.

Logan Welker has 18 years of experience in the agriculture industry, including 4 years working with the USDA's Natural Resources Conservation Service (NRCS) and 12 years working in ag retail. Welker holds a B.A. in Ag Business and an M.S. in Agronomy from Iowa State University. Welker has significant experience in the agriculture industry, including working in areas related to grazing management, cover crops, grain quality, soil science, and weed management. His specialty areas are in soil fertility and ag technology for regions of the country where corn, soybean, and wheat are the predominant crops.

Turnstile 5: Agronomically feasible values (L4T5) Turnstile 5 confirms that certain data attributes represent agronomically feasible values. Indigo's subject matter experts defined the ranges in L4T5. These values represent extreme upper bounds to what is agronomically possible. These checks are applied in the Carbon by Indigo Web UI when growers enter their data, as well as in the QA / QC pipeline in L4T5 as a second confirmation that values are agronomically feasible.

Table 13. Agronomically feasible ranges for yield.

Crop	Bu / Acre	Kg / Acre
Alfalfa	None	0 / 20,000
Barley	0 / 225	0 / 4,905
Beans/Peas	0 / 67	0 / 1,814
Canola	0 / 100	0 / 2,268
Corn	0 / 2,000	0 / 50,848
Cotton	0 / 8	0 / 1,814
Flax	0 / 75	0 / 1,905
Oats	0 / 200	0 / 2,900
Peanuts	0 / 779	0 / 6,000
Rice	0 / 300	0 / 6,120
Rye	0 / 150	0 / 3,810
Sorghum	0 / 225	0 / 5,715
Soybeans	0 / 175	0 / 4,760
Sugar Beets	0 / 2,325	0 / 40,000
Sunflowers	0 / 125	0 / 1,361
Triticale	0 / 225	0 / 6,120
Wheat	0 / 225	0 / 6,120

Table 14. Agronomically feasible ranges for rate attributes.

Event Attribute	Range [unit]
Organic amendment application rate	0 / 400,000 [kg / ha] 0 / 400,000 [bu /ha[
Chemical application N application rate	0 / 500 [kg / ha]
Chemical application N application percent for custom product	0 / 46 [%]

Summary of L4 source data and turnstile checks With the degree of complexity and interrelationships among the various Level 4 checks, the following table outlines the quantitative (and also qualitative for negative event) reviews performed across all relevant events and associated attributes. Recall that "negative" events are grower attestations that a management practice did *not* occur, and for some events, we have reviewed this qualitative information. (We also check against crop type (also qualitative) because it is fundamental to our project definition).

As many of the data sources used for these checks are (1) based on internal, calibrated models or (2) provided by a third party, there are situations where, for certain crops, regions, etc., there is no returned source data from NASS or remote sensing to perform the associated check. When that occurs, we either rely on another turnstile or, in rare cases, fall back to a similar check from a lower level (e.g., L2). L2 backup checks are indicated in Table 15. For details about the data sources used in Turnstiles 2 – 4, please see the relevant sections above and/or IndigoCarbon_US-1_2023_0069.

Table 15. Summary of check(s) applied to each attribute.

Event Type	Attribute	L4T2	L4T3	L4T4	L4T5	L2 Backup
Cash Crop Planting	Crop *	CDL				
	Planting date		NASS			Y
Cover Crop Planting	Cover crop species *	CDL				
	Cover crop planting date	RS **				Y
	Harvest date	RS **	NASS			Y
Harvest	Cover crop harvest date	RS				Y
	Yield		NASS	General Crops	Y	
	Actual production yield		NASS	Crops		
Termination	Termination date	RS	NASS			Y
	Cover crop termination date	RS		General		Y
Tillage	Tillage date	RS **,***				Y
	Tillage depth			Crops		
Irrigation	Irrigation date	RS **		General		Y
Chemical Applications	N application rate		NASS	Crops	Y	
	Application blend percent			Crops		
	N application percent		CDMS	General Crops		
	P_2O_5 application percent		CDMS	General Crops		
	K_2O application percent		CDMS	General Crops		
	S application percent		CDMS	Crops		
	Anhydrous application rate			General		
	Dry fertilizer application rate			General		
	Liquid fertilizer application rate			General		
	Application date			General		
	Application rate		ARMS	Crops	Y	
Organic Amendments	Liquid manure application rate			General		
	Solid manure application rate			General Crops		
	Application date			General		

* If the grower could not provide documentation, or if their provided documentation is insufficient to address the value-out-of-range error during evidence review, we relied on the grower attestation

provided at final filing as evidence of the management practice. We used the grower-submitted qualitative (i.e., attested) value and did not replace the value with an extrapolated or default value. (See Gate 1 above for details on final filing.)

** Remote sensing (RS) algorithms are used to review both positive (occurred) and negative (did not occur) events. See Turnstile 2 above for more details.

*** For negative tillage events that have no checks because of missing source data or that conflict with RS tillage checks, no backup process was implemented. We relied on grower attestation during final filing for qualitative event confirmation. (See Gate 1 above for details on final filing.)

1.2.7 Level 5: Additionality assessment (L5)

Once all of the data have been reviewed and determined "of quality," we will use the events and attributes to assess additionality, following the additionality assessment process described in IndigoCarbon_US-1_2023_0018, and using the "project-specific measures for additionality" defined in IndigoCarbon_US-1_2023_0024a. The L5 assessment is a key component of meeting SEP v1.1 Performance Standard Test (PST) for additionality, as this check reviews all grower-entered event data and determines if a grower will fulfill the additionality condition for any practice change (not just the one indicated). Here we note "grower-entered event data" to indicate that no gapfilled data of any kind – extrapolated or default values – are used in the initial L5 additionality assessment. Growers must be additional all on their own, or they are not allowed to pass L5 and final file.

Criteria for and determination of additionality are available in IndigoCarbon_US-1_2023_0018. The additionality assessment is performed using the historical management period by applying similar concepts used to construct baseline threads for modeling. For L5 all categories of practice changes use an approach emulating "blended" baselines (per SEP v1.1 Section 3.4.1.3) to determine additionality, with the exception of the "nitrogen application" category of practice changes that employs an approach emulating a "matched" baseline, as nitrogen application is often highly crop-specific. (More information on these different approaches is available in IndigoCarbon_US-1_2023_0048.) A quarter (25%) of the field has to be additional to pass L5, and fields that do not qualify for additionality cannot be final filed. (See Gate 1 above for details on final filing.)

After additionality checks have screened fields for qualifying practice changes, L5 flags are used to identify what can and cannot be credited for various reasons. The most important of these flags indicates creditability (i.e., indicates whether or not a field should be modeled and processed for CAR credits using the selected biogeochemical model). Even though a field may be *additional* according to internal Indigo checks, the field might not be *creditable* because it undergoes deep-till or it exhibits common practices (i.e., CAR Negative List). An L5 pass is required for a grower to final file.

1.2.8 Level 6: Gapfillability

"Gapfillability" is the capacity of an attribute or event to be gapfilled. Recall that "gapfilling" is the replacement of missing (or out-of-range) parameters with defaults based on survey data (or other SEP-approved sources, following the SEP v1.1 6.1 data hierarchy (2)), or through extrapolation from a grower's existing data, following concepts described for in SEP v1.1 6.1 data hierarchy (1c). L6 evaluates if a missing (or out-of-range) attribute (or whole event!) could be gapfilled. Given the heavy data burden on Field Managers, some data validation errors can be resolved by Indigo (on behalf of the grower). L6 is a type of "override" for certain lower level errors (as noted in the

following table), so growers can focus on the most critical ones, while leaving the rest for Indigo to address for them.

Table 17. Errors resolvable by gapfill approach.

Gapfill approach	Use on practice change data	Errors resolvable by gapfilling
Data extrapolation	• With-project period – No • Historic period – Yes, but only from other baseline events in the same field that qualify *	• L1 missing events • L1 missing attributes
Default replacement	No	• L1 missing events • L1 missing attributes • L4 value out-of-range

* See IndigoCarbon_US-1_2023_0031.

Note that gapfillability interacts with the "declared practice change" designation to restrict how missing attributes or events pass through the Carbon data pipeline (Table 17). This relationship is intended to place safeguards on the practice-changing data that are most critical for additionality and credit quantification. These safeguards, in combination with the L5 rerun after gapfilling is complete (described in the previous section), ensure that gapfilling does not allow growers to enter the program and be credited who should not.

1.3 Gate 3: Evidence review & confirmation

The evidence review processes aim to reduce the risk of a material misstatement, namely that we do not overestimate our greenhouse gas emission reductions (i.e., carbon credits) by over 1%. We have designed the following protocol to ensure program eligibility (i.e., enrolled fields are eligible to participate) and to maximize the quality of data going into the biogeochemical model. We make meaningful effort to ensure the quality of information in our project by checking 100% of evidence associated with out-of-range (outlier) quantitative data and as much additional corroborating information that growers provide as possible.

1.3.1 Selecting evidence for review

The following table shows the rates at which evidence is sampled for various types of grower-entered information. Any attribute that is found to be out-of-range requires evidential proof (of the kind noted in IndigoCarbon_US-1_2023_0029) for those data to be used in their unaltered form. A certain subset of otherwise passing attributes may be randomly selected for follow-up (secondary) review (similar to TSA's random "extra" screening). Out-of-range failures show up in the Carbon by Indigo UI (Gate 1) as errors, and the grower will have the option to remedy them by changing the values or adding (more) evidence. In cases where we are able to gapfill, they are more like warnings (than errors), which growers can choose to have Indigo resolve for them. Note that *only* out-of-range, non-gapfillable errors require evidential proof from the grower; all other evidence is *requested* but not required.

Table 18. List of evidence sample rates for different categories of variables. For more information about which attributes are qualitative vs quantitative, see IndigoCarbon_US-1_2023_0029.

Variable Category	Quantitative Out-of-Range	In-Range	Qualitative *
Practice Change	100% **	100%	100%
Sensitive (see Table 19)	100% **	2.5%	2.5%

* Includes quantitative dropdowns. ** Unless there is currently evidence in review.

Table 19. List of event attributes that are tracked as practice changing and/or sensitive and are marked for evidence upload and review.

Event Type	Event Attribute	Practice Changing	Sensitive *
Positive Events			
Cash Crop Planting	Crop Type	X	X
	Planting Date	X	X
Cover Crop Planting	Cover Crop Species	X	X
	Cover Crop Planting Date	X	X
Tillage	Tillage Date	X	X
	Tillage Method	X	X
Chemical Application (fertilizer)	Product Rate	X	X
	% N in Product	X	X
	Product Form (of Nitrogen)	X	X
Organic Amendment (manure)	Amendment Type	X	X
	Application Rate	X	X
Irrigation (Summary)	Irrigation Date		X
Harvest/ Termination Event	Crop (Type or Species)	X	X
	Harvest Date	X	X
	Termination Date	X	
	Harvest Method	X	X
	Harvest Yield	X	X
	Residue Burned	X	X
Negative Event			
Cash Crop Planting	Planting Date	X	X
Cover Crop Planting	Cover Crop Planting Date	X	X
Tillage	Tillage Date	X	X
Chemical Application (fertilizer)	Application Date	X	X
Irrigation (Summary)	Irrigation Date		X

* Sensitive variables are those noted as DayCent-CR v1.0.2 "1st tier priority" and whose correct entry will have a major impact on model results. For this reason we make meaningful effort to ensure that these associated attributes are accurate and of quality. See IndigoCarbon_US-1_2023_0072 for more information.

1.3.2 Using the hierarchy of data sources as evidence

As outlined in Section 6.1 of the SEP v1.1, our guiding principle and overarching assumption is that data from the top of the hierarchy are more valuable to our project overall than values sourced lower

on the hierarchy. More specifically, we might say that data classified as (1a): historical management records are "higher integrity" than (1c): signed attestation from the Field Manager. Of specific relevance to the concept of evidence are the following types of historical management records (e.g., logs or files containing machine and/or sensor data), and remote sensing (e.g., satellite imagery, manned aerial vehicle footage, or drone imagery). Growers do not need to provide documentation for attributes tagged as requesting or requiring evidence if these data fall into one of two (1a) categories.

(i) Logs or files containing machine and/or sensor data: As noted above in Gate 1, in addition to manual data entry, we expanded the Carbon by Indigo software to include data ingestion from MyJohnDeere. Since these are machine data, they are considered "self-corroborating" and no further action need be taken.

(ii) Remote Sensing: Indigo's remote sensing results can serve as evidence for the presence/absence of four events and one quantitative attribute: cover crops, tillage, irrigation, and harvest (harvest date). For cases where corroborating evidence is requested or required *and* the error is *not* associated with a remote sensing check, remote sensing can serve as evidence and no further action need be taken. Note that these results are also used in Gate 5 for gapfilling by default replacement (IndigoCarbon_US-1_2023_0031).

1.3.3 Reviewing and confirming manually uploaded evidence

The Evidence Review Team (ERT) uses a rigorous process to determine if the documentation submitted for review sufficiently substantiates the grower-provided values, or if the evidence needs to be passed back to the grower for adjustment. Generally, provided evidence needs to be field and date specific. Document types accepted are per event type and informed by subject matter experts. Fig. 2 outlines the criteria used by Indigo to evaluate various forms of evidence submitted by the Field Managers.

Figure 2. Criteria used to by the Evidence Review Team to evaluate evidence submitted by Field Managers. The system flags both positive and negative events* for evidence review, and the evidence criteria noted below are applicable to both.

		ACCEPTABLE SUPPORTING DOCUMENTATION (ANY OF THE BELOW)							
EVENT	Practice Information	Farm Mgmt. Software or Machine Data Reports	FSA 578 or Crop Insurance Reports	Application Invoices or Fertilizer Receipts	Production Report	Scale tickets and transaction receipts	In-field Photos (Geo-, Time-Stamped)	NRCS Reports	Other Mgmt. Records or Plans
Planting	• Crop Type • Planting Date	✓	✓				✓		✓
Fertilizer	• Fertilizer Type • Application Date • Application Rate • Application Method • Inhibitor	✓		✓				✓	✓
Harvest	• Yield • APH Yield • Harvest Date • Harvest Method (Residue Management) • Failed Crop Date	✓	✓		✓	✓			✓
Tillage	• Tillage Date • Tillage Equipment	✓					✓	✓	✓

		ACCEPTABLE SUPPORTING DOCUMENTATION (ANY OF THE BELOW)						
EVENT	Practice Information	Farm Mgmt. Software or Machine Data Reports	Application Reports or Fertilizer Receipts	FSA 578 or Crop Insurance Reports	Seed Invoices	In-field Photos (Geo-, Time-Stamped)	NRCS Reports or Feed Receipts	Other Mgmt. Records or Plans
Organic Amendment (If Applicable)	• Organic Amendment Type • Application Date • Application Rate • Application Method	✓	✓					✓
Cover Crop (If Applicable)	• Cover Crop Type • Cover Crop Planting Date • Harvest/Termination Date • Harvest/Termination Method	✓		✓	✓	✓	✓	✓
Grazing (If Applicable)	• Animal Type • Grazing Start/End Dates • Stocking Rate • Grazing Method • Grazing Mgmt. Plan • Supplemental Feed	✓				✓	✓	✓
Irrigation (If Applicable)	• Crop Irrigated (Y/N) • Irrigation Method • Irrigation Start/End Dates • Flood/Drain Dates (Rice Only)	✓				✓		✓

Each form of evidence was manually reviewed by Indigo using the administrative panel of the agronomy data service (ADS). Any evidence of equipment that is provided through a photo will be reviewed to ensure that all equipment appears to be operating in alignment with manufacturer specifications. Upon completion, Indigo assigned statuses to each document type to indicate whether it was an accepted source to substantiate the grower-provided values. The statuses assigned, along with the following actions required to achieve resolution, are detailed in Table 20.

Table 20. Description of documentation statuses and required actions (if applicable).

Document Type Status	Document Type Status Description	Error message displayed in the IC UI
PASS	Evidence matches value and is sufficient	
EVIDENCE_ERROR	Evidence is not relevant and, if required, should be replaced	FLAGGED DATA Provided documentation does not demonstrate <event attribute(s)>.
EVIDENCE_INSUFFICIENT	Evidence is relevant but needs to be supplemented with additional evidence, or evidence is not provided at all	FLAGGED DATA More documentation is needed for <event attribute(s)>.
DATA_ERROR	Evidence is good, but data do not match, and the value should be changed to match the evidence	FLAGGED DATA Provided documentation does not demonstrate <event attribute(s)>.
IN_PROGRESS	Evidence does not fit an existing category and requires additional review	
GAPFILL	Value should be gapfilled because it is out-of-range and not a critical question	

Document types that are assigned as PASS or GAPFILL are accepted as valid sources of evidence and allow that field to continue through the Carbon data pipeline. However, the remaining statuses (i.e., EVIDENCE_ERROR, EVIDENCE_INSUFFICIENT, and DATA_ERROR) require further action from the Field Manager. Any time a document is flagged as such, the following request is submitted:

> *Attach supporting document(s) that demonstrates this <event attribute> information. If entered value is incorrect, update information.*
>
> *Documentation should clearly display <event attributes>, field name and/or boundary (if available), and event date (if available).*

Once the Field Manager resubmits documentation, the evidence review process is restarted. Note that throughout this process, data submitted from the Field Managers are never altered, though values may be replaced through gapfilling with defaults (*after* final filing), and stored separately in the system.

Since the evidence review process is entirely manual, that leaves the opportunity for review mishaps, however small. In instances where evidence was mishandled (i.e., approved as corroborative but should not have been), there is a backup process to reaffirm corroboration for extreme outliers and to override as "EVIDENCE_INSUFFICIENT" in those instances. For this narrow edge case, these data are gapfilled where applicable or excluded from the reporting period.

1.4 Gate 4: Grower outreach & follow-up

This gate acts as a buffer between the results determined in Gates 2 and 3 and the actual grower (for results that are not automatically shared back to the grower through the Carbon by Indigo UI). We do not want the grower to be contacted by several different CCS team members, each with their own set of questions and needs. Instead these questions, needs, and follow-ups are "pooled" until a certain "outreach threshold" is reached, and then one specific CCS teammate, usually the one most familiar with that grower, reaches out to address a number of known issues. They then work specifically with the grower to address mistakes, mismatches, missing data, etc.

If new or missing data are entered (at any time), that specific field re-enters the Carbon data pipeline from the beginning (Gate 1), and cycles through Gates 1 – 3(4) as many times as needed until either one of the following becomes true. From there, all data flow to Gate 5: Gapfilling (IndigoCarbon_US-1_2023_0031).

- All needed details have been entered, they pass all quality and evidence checks in Gates 2 and 3, and the field is ready to final file. (See Gate 1 above for details on final filing.)
- We have exhausted our communication options in Gate 4, and will accepted that what data currently exist are all that we will get, knowing that some growers may be be unable to final file from missing (or incorrect) data.

2 Summary of Changes

Section	Date	Summary of Changes
Level 4: Data confidence (L4)	June 26, 2024	Added QC ranges for millet